

SQL Sales Data Analysis Report

This report presents SQL-based analysis performed on a sales dataset containing transactional information related to products, payment methods, referral sources, coupon usage, customer purchases, and revenue performance.

1. Project Objective

The objective of this project was to use SQL queries to extract business insights, analyze sales performance, identify customer purchasing trends, and evaluate revenue distribution across products, payment methods, referral sources, and promotional coupons.

2. Tools Used

- SQLite
- DB Browser for SQLite

3. SQL Concepts Used

- SELECT
- WHERE
- GROUP BY
- ORDER BY
- COUNT()
- SUM()
- AVG()
- ROUND()
- LIMIT

4. Total Revenue Analysis

The SQL analysis revealed total sales revenue of \$1,264,761.96, indicating strong transactional performance and consistent customer purchasing activity across the dataset.

5. Revenue by Product

Chair products generated the highest revenue at \$195,620.11, followed closely by Printers at \$195,612.61 and Laptops at \$192,126.56. Phones generated the lowest product revenue contribution at \$151,722.39.

6. Orders by Payment Method

Online payments recorded the highest number of orders with 258 transactions, followed by Cash with 246 orders and Credit Card payments with 234 orders. Customer payment preferences were relatively balanced across all payment methods.

7. Top Customers by Revenue

Customer C38840 generated the highest customer revenue at \$5,723.23 across 2 orders. Several additional customers generated over \$3,300 in revenue, indicating the presence of high-value customer transactions.

8. Order Status Analysis

Cancelled orders recorded the highest order count at 250 transactions, followed by Returned orders at 247 and Pending orders at 237. Delivered orders represented the lowest count at 231 transactions.

9. Referral Source Analysis

Instagram generated the highest referral revenue at \$275,285.45, followed by Email at \$261,808.55 and Google at \$250,441.48. Referral traffic produced the lowest revenue contribution at \$226,815.58.

10. Revenue by Payment Method

Credit Card transactions generated the highest revenue at \$263,847.63, followed closely by Online payments at \$262,442.94 and Cash payments at \$259,786.29. Debit Card transactions produced the lowest payment revenue contribution.

11. Average Order Value Analysis

Credit Card payments generated the highest average order value at \$1,127.55. Debit Card transactions produced the lowest average order value at \$1,001.56, indicating moderate variation in customer spending behavior.

12. Coupon Performance Analysis

The FREESHIP coupon generated the highest revenue at \$335,036.99 across 313 orders. NO COUPON transactions followed with \$322,401.41 in revenue, while SAVE10 and WINTER15 generated similar revenue contributions.

13. Conclusion

The SQL analysis successfully identified important sales trends, customer purchasing patterns, payment preferences, referral performance, and coupon effectiveness. The project demonstrates practical SQL querying, data aggregation, and business intelligence analysis skills.